👑 CODERKING STUDIO

Official Technical Roadmaps & Architecture Guides

HASH RING

System Design: Consistent Hashing & Distributed Ring

Jab cluster mein ek naya server add hota hai, toh pura cache crash huye bina kaise bachta hai? 🏢🌐

1. 360° Hash Ring Space

Har physical server ko multiple virtual points par hash karke ring par balance spread kiya jaata hai.

```
# VIRTUAL NODES
हर physical server को multiple virtual
points पर hash करके ring पर balance spread
किया जाता है।
```

2. Virtual Node Tokens

Server add ya remove hone par sirf adjacent neighbour keys move hoti hain, baaki sara cluster unaffected rehta hai.

```
# ZERO REHASH
Server add या remove होने पर सिर्फ adjacent
neighbour keys move होती हैं, बाकी सारा cluster
unaffected रहता है।
```

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Official GitHub: **github.com/shubh74/coderking-roadmaps**